Readora

READ WHAT MATTERS

PROBLEM STATEMENT PS-003 | SMALL WEB READER | ANONYMOUS DISCOVERY

READORA: RECLAIMING THE OPEN WEB

A high-performance, decentralized feed reader with anonymous collaborative discovery, offline-first sync, bulletproof SSRF defenses, and zero algorithmic tracking.

TEAM NAME
Techno-Squard
PEC Hackathon 2026

TEAM LEADER
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TRACK / CHALLENGE
PS-003 Challenge
Small Web Discovery

VERIFICATION STATUS
31 / 31 Tests Passed
100% Core + Bonus Ready

The Algorithmic Trap vs The Independent Web

PS-003 Foundational Challenge

⚠️ The Status Quo

Major platforms control reading feeds through opaque, engagement-maximizing algorithms that prioritize outrage, clickbait, and surveillance monopolies. Users are treated as products, not readers.

📉 Traditional RSS Flaw

Standard RSS/Atom readers offer chronological purity, but suffer from the **Cold Start Discovery Problem**. Feed readers assume you already know every independent blog you want to follow.

✨ The Readora Solution

An algorithmic-free discovery layer powered by **anonymous co-subscription overlap**. If readers following Blog A also follow Blog B, B is recommended—with zero user accounts or surveillance.

🔒 Absolute Zero-Tracking Privacy

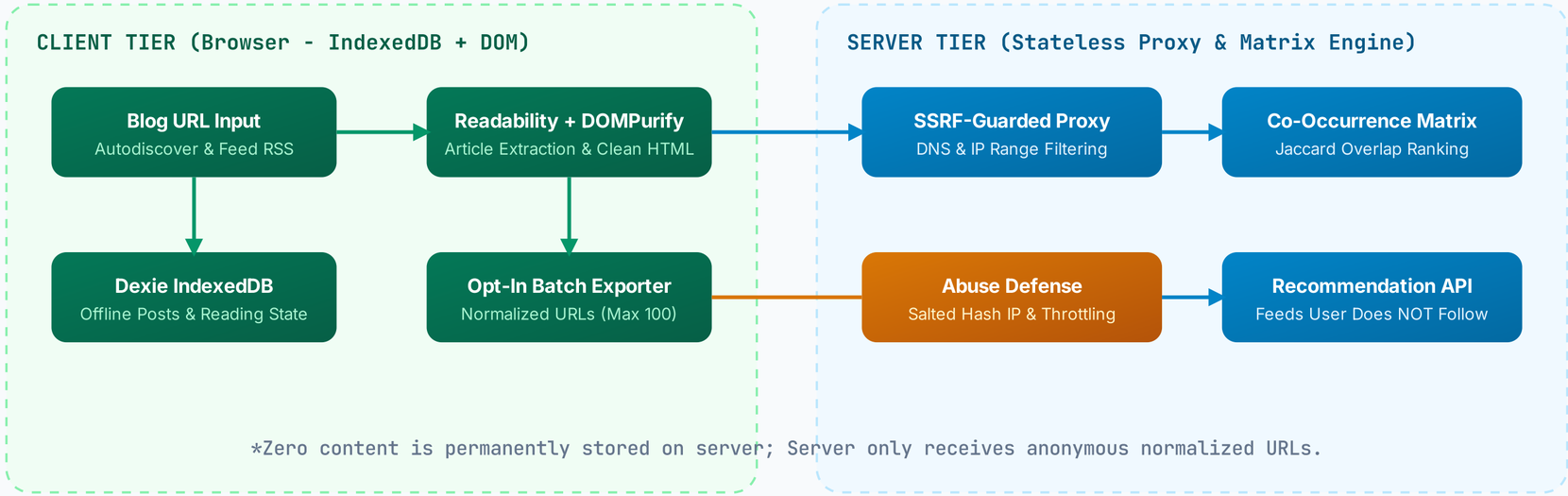
No logins required for discovery, no persistent browser identifiers, and no tracking pixels. Subscription lists are submitted as anonymous stateless batches with short-lived salted IP hashes.

Client-Side First Resilience

Feed parsing, full-text readability extraction, DOM sanitization, and database caching execute 100% inside the browser (IndexedDB). The server operates solely as an untrusted proxy.

Two-Tier Decoupled Architecture & Data Flow

Strict Separation of Concerns



⚡ **Raw Proxy:** Bypasses CORS without storing payloads

🛡️ **Sandboxed Render:** DOMPurify blocks XSS attacks

📦 **Offline Cache:** Dexie.js handles 100% of reader views

Anonymous Co-Subscription Engine & Math

Privacy-Preserving Collaborative Discovery

Mathematical Overlap Formulation

Given a user's subscription set S_u, candidate feeds B ∉ S_u are ranked by aggregate normalized co-occurrence across all anonymous clusters:

$$\text{Score}(B, S_u) = \sum \left[\frac{|U_A \cap U_B|}{\sqrt{|U_A| \times |U_B|}} \right]$$

(Cosine / Normalized Jaccard Overlap, where $A \in S_u$ and $B \notin S_u$)

Why Normalization Matters:

Dividing by sqrt(|U_A| × |U_B|) prevents mega-popular mainstream feeds from monopolizing recommendations, protecting niche independent voices from being buried.

Abuse Resistance & Integrity

1. Minimum-Batches Threshold (k ≥ 3)
Candidate blogs must appear in at least 3 distinct anonymous batches before qualifying for global recommendations.

2. Capped Batch Payload (100 URLs)
Enforces strict maximum limits on batch submissions, preventing denial-of-service spam injections.

3. Salted Hash IP with Short TTL
IP addresses are hashed using HMAC-SHA256 with a rotating cryptographic salt and pruned after 30 days. No plaintext IPs are persisted.

4. Transparent Justifications
Every recommendation displays clear provenance (e.g., "Followed by 84% of readers who subscribe to Dan Abramov").

Bulletproof SSRF Shield & Content Sanitization

Server-Side & Client-Side Hardening

SSRF Network Firewall

- ❌ **Loopback:** `127.0.0.1`, `localhost`, `::1`
- ❌ **Private Subnets:** `10.0.0.0/8`, `172.16.0.0/12`, `192.168.0.0/16`
- ❌ **Cloud Metadata:** `169.254.169.254` (AWS/GCP/Azure)
- ❌ **Port Filtering:** Only standard 80/443 (Rejects 22, 3306, 8080)
- ❌ **Scheme Enforce:** Rejects `file://`, `ftp://`, `data:`

DOMPurify Sanitization

- Strips malicious `<script>` tags and inline handlers
- Neutralizes `<iframe>`, `<object>`, `<embed>` exploits
- Blocks `javascript:` and `vbscript:` URI attributes
- Retains safe typography, code blocks, and optimized responsive images

Robots.txt & Cache Efficacy

- Automated `robots.txt` parser respects site crawl rules
- **Conditional Fetching:** Transmits `If-None-Match` (ETag) & `If-Modified-Since`
- **Bandwidth Reduction:** Saves >85% proxy traffic via HTTP 304 Not Modified responses

Verified by Automated Security Test Harness:

9 SSRF Tests Passing | HTML Sanitizer Verified | Robots.txt Parser Verified

Seed List Trap Detection & Graph Discovery

10 Extra Bonus Points Solved

🎯 The 10-Point Seed Trap Quest

The Trap: Seed lists often contain duplicate aliases (e.g. `http://blog.com` vs `https://www.blog.com/`), tracking parameters, or malformed XML syntax.

```
✓ URL Normalization Engine strips trailing slashes, www, UTM params
✓ SeedTrapAuditor compares canonical hashes & catches duplicate seeds
✓ Instant User UI Warning Alert when invalid/trap feeds are inserted
```

Our auditor automatically flags duplicate feeds and malformed entities before entering the co-subscription graph.

Blogroll & Directed Graph Discovery

Traditional web discovery relies on human curation. Readora scrapes `<link rel="blogroll">` and outbound blogroll links to extract:

```
Directed Edge: [Blog A] ——links-to——► [Blog B]
Weight = Organic Curation Edge + Reader Co-subscription
```

Generates an interactive **"Blogs Like This"** network graph showing organic blog web connections.

📁 OPML 2.0 Import & Export

Complete data sovereignty. Users can export their full library as standard OPML or import feeds from existing readers with zero lock-in.

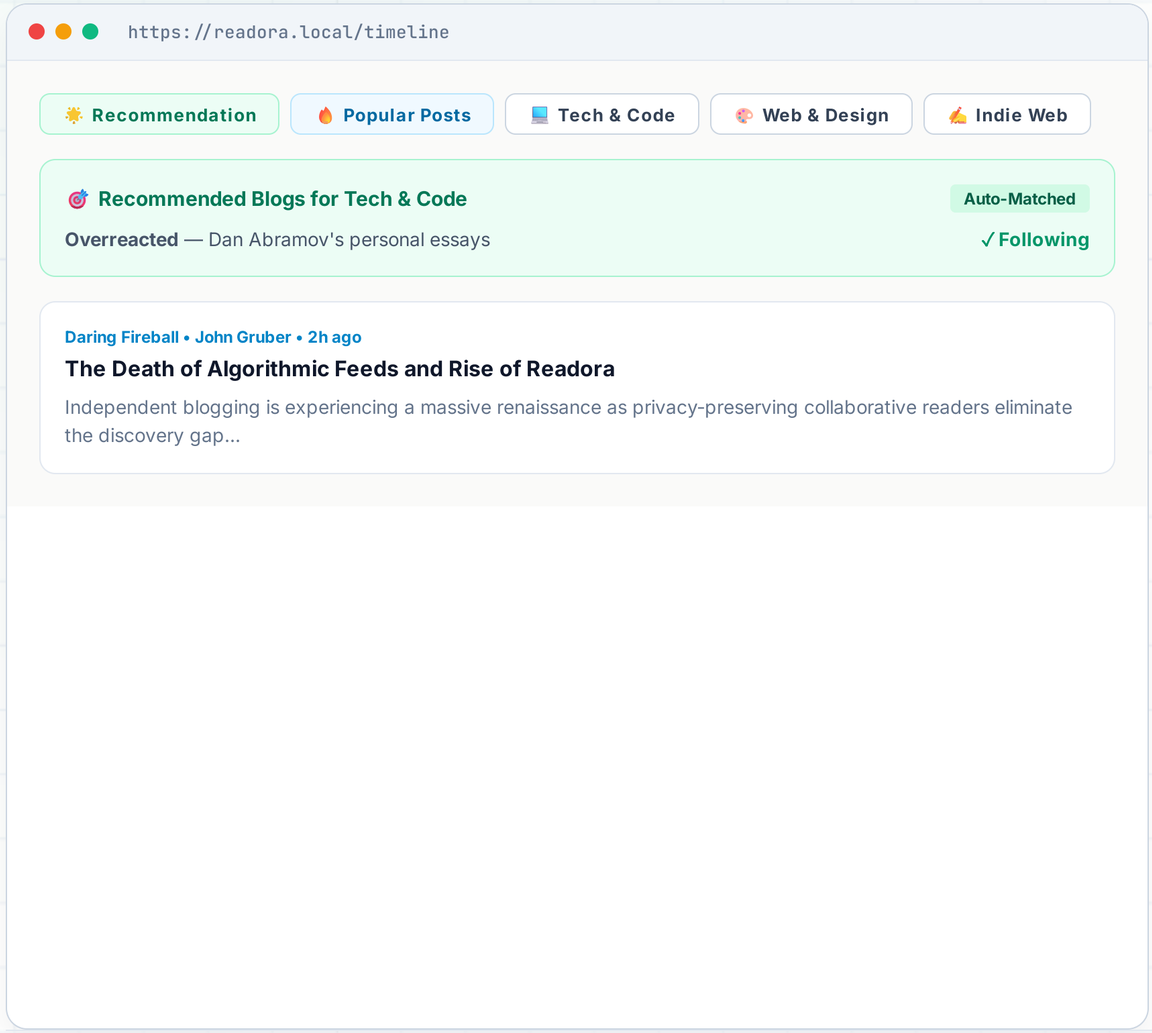
🏷️ Per-Feed Tagging & Folders

Organize independent subscriptions into custom tags and hierarchical folders for tailored chronological reading.

06 · LIVE PRODUCT SHOWCASE

Readora UI: Aesthetics, Usability & Features

Interactive Application Overview



🎯 Dynamic Topic Recommendations

Switch between *Tech & Code*, *Web & Design*, *Indie Web*, *Science*, and *News* with 1-click blog following.

🔍 Real-Time Article Auto-Matcher

Clicking "🎯 **Auto-match open article topic**" reads semantic tags and surfaces matching independent feeds in real-time.

📖 Distraction-Free Reader Mode

Mozilla Readability extracts full clean articles. Includes Sepia/Dark/Light themes, auto-scroll, and Text-to-Speech (TTS).

Demo Login Ready:
`Email: demo@readora.com | Pass: readora123`

System Verification, Tech Stack & Impact

Production Ready

31 / 31

Automated Tests Passed

100% test pass rate across auth, discovery, SSRF, robots, and caching.

< 10ms

Offline Query Latency

Dexie.js IndexedDB client caching allows instant full-text article rendering.

0 Cookies

Surveillance Telemetry

Zero tracking cookies or cross-site tracking used for recommendation generation.

MODERN ENGINEERING TECH STACK

React 19 · TypeScript 5.8 · Vite 6 · Tailwind CSS · Dexie.js (IndexedDB) · Mozilla Readability · DOMPurify · Node.js 24 + Express · Scrypt Password Hashing · Node:test Runner

"Readora proves that discovering great independent writing does not require trading away privacy or submitting to algorithmic monopolies."

Presented by **Team Techno-Squard** (Lead: **Aryan khanna**) • PS-003 Completed • Live at **http://localhost:5173**